

PROJECT APEX

Enterprise LNG Operations Intelligence Platform

VOLUME 1

Project Planning & Business Understanding

An End-to-End Power BI Analytics Project

PRESENTED AS A SIMULATION OF

Atlantic LNG Operations Limited (ALOL)

A Fictional LNG Operations Company Modeled on Nigeria LNG (Bonny Island)

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Series Overview

Volume 1 — Project Planning & Business Understanding (this document)

Volume 2 — Data Preparation & Data Modeling

Volume 3 — DAX Development

Volume 4 — Dashboard Design & Storytelling

Volume 5 — Portfolio, Documentation & Interview Preparation

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1. Project Background

Liquefied Natural Gas (LNG) is one of the most strategically important commodities in global energy trade. Nigeria, sitting on the ninth-largest proven natural gas reserves in the world, has built its LNG export economy around a single, dominant facility: Nigeria LNG Limited (ALOL), located on Bonny Island in Rivers State. ALOL has operated continuously since 1999, growing from two production trains to six, and currently constructing a seventh (Train 7) to expand capacity from 22 million tonnes per annum (MTPA) toward 30 MTPA.

This project does not use ALOL's real operational data — that data is proprietary and confidential. Instead, this project simulates a fictional company, Atlantic LNG Operations Limited (ALOL), an LNG operator modeled closely on ALOL's real-world structure: six operational liquefaction trains plus a seventh under construction, multiple product streams (LNG, LPG, Condensate, NGL), a global customer base spanning Europe, Asia, the Middle East, the Americas, and Africa, and the full operational footprint of a major industrial gas exporter — production, exports, maintenance, shutdowns, safety incidents, finance, and weather.

The purpose of building a fictional but realistic dataset is to allow the development of a complete, portfolio-grade Power BI analytics solution without any confidentiality, licensing, or data-protection constraints, while still requiring the same rigor, domain literacy, and technical sophistication that a real oil & gas analytics engagement would demand.

1.1 Why This Project Exists

This project was initiated to build a demonstrable, professional-grade Power BI portfolio piece that proves competency across the full business intelligence lifecycle: requirements gathering, data modeling, Power Query transformation, DAX measure development, dashboard UX design, and executive storytelling. Rather than working with a generic retail or sales dataset (the most common portfolio choice), this project intentionally targets a complex industrial domain — oil & gas / LNG operations — because it forces engagement with:

- Multi-grain fact tables (daily production vs. per-shipment vs. per-incident)
- Time intelligence across both calendar and financial periods
- Operational KPIs unique to process industries (availability, utilization, OEE-style metrics)
- Geographic/map-based analysis of global trade flows
- Safety and compliance reporting (HSE), which is mandatory in real energy companies
- Finance-to-operations linkage (how uptime and incidents affect profitability)

1.2 Project Type

This is a self-directed, simulated business intelligence consulting engagement. The working assumption throughout all five volumes is that Atlantic LNG Operations Limited (ALOL) has commissioned an

external/internal BI analyst (the learner completing this project) to design and deliver a Power BI reporting suite covering its core operations. All documentation in this and subsequent volumes is written in the same register and structure a paid consulting engagement would use, so that the finished work doubles as portfolio evidence of professional practice, not just a personal practice exercise.

1.3 Relationship to the Dataset Already Built

The underlying dataset for this project has already been engineered as a proper star schema and was delivered ahead of this document as a set of CSV files. It consists of 7 dimension tables and 7 fact tables totalling approximately 143,000 records, covering the period 1 January 2019 to 31 December 2024 (a six-year operating history). Section 6 (Dataset Overview) and Section 7 (Data Dictionary Summary) of this volume describe that dataset in full. Volume 2 will take this dataset through Power Query cleaning, validation, and modeling inside Power BI Desktop.

2. Business Context

The global LNG market has entered a period of structural growth and structural risk simultaneously. Understanding this context is essential before any dashboard or metric is designed, because dashboards exist to help real people make real decisions inside this market environment.

2.1 The Global LNG Market

LNG demand has grown steadily as countries seek a lower-carbon alternative to coal and oil, and as energy security concerns (particularly in Europe following supply disruption from traditional pipeline sources) have pushed buyers toward diversified, ship-borne gas supply. Asia (Japan, South Korea, China, India) remains the largest demand region, but Europe has become an increasingly aggressive buyer of spot and short-term cargoes. This has direct implications for how an LNG producer like Atlantic LNG Operations Limited must operate:

- Production reliability is commercially critical — a single train going offline unexpectedly can cost millions of dollars in lost cargoes and contractual penalties
- Customers expect a mix of long-term Sale and Purchase Agreements (SPAs) for revenue stability and spot/short-term sales for margin capture
- Shipping and logistics (vessel scheduling, jetty loading windows, freight cost) materially affect realized margin per cargo
- Planned maintenance (turnarounds) must be scheduled to minimize overlap with peak demand and contractual delivery windows

2.2 The Nigerian Operating Environment

Operating an LNG facility in Nigeria carries a distinct set of contextual factors that any realistic simulation must reflect:

- Feed gas supply risk — pipeline vandalism, third-party interference, and upstream production issues in the Niger Delta can interrupt the gas supply that feeds the liquefaction trains
- Tropical climate — Bonny Island's high humidity, heavy rainy-season rainfall, and Harmattan dust haze affect both plant operations and marine/shipping conditions
- Regulatory and fiscal obligations — Petroleum Profits Tax (PPT), royalties, and Nigerian content requirements shape both the finance function and the workforce composition
- Community and security considerations — coastal/riverine operations require constant HSE vigilance around marine operations and community relations

2.3 Why Operational Analytics Matters Here

In a capital-intensive, continuous-process industry like LNG, the gap between a top-quartile operator and an average one is almost entirely a function of two things: production uptime (availability and utilization) and unplanned event management (shutdowns, maintenance failures, HSE incidents). A 1% improvement in plant availability across six trains translates into millions of dollars of additional revenue per year, because every unit of feed gas not converted to LNG due to downtime is a direct, unrecoverable revenue loss — there is no inventory buffer the way there is in manufacturing.

This is precisely why this project's data model and KPI plan (defined fully in Volume 3) center on availability, utilization, and downtime, rather than purely financial metrics. Finance dashboards alone would miss the underlying operational drivers; production dashboards alone would miss the commercial consequences. The project is deliberately designed to connect the two.

3. Company Profile (Fictional LNG Company)

“Atlantic LNG Operations Limited (ALOL) is a fictional company created exclusively for this educational and portfolio project. Any resemblance to the operations of Nigeria LNG Limited is intentional in terms of industrial structure and process design, but all figures, names, and records are entirely synthetic.”

3.1 Company Overview

Attribute	Detail
Company Name	Atlantic LNG Operations Limited (ALOL)
Industry	Liquefied Natural Gas (LNG) production and export
Headquarters	Bonny Island, Rivers State, Nigeria
Incorporated	Fictional — modeled on a company founded 1989, operational since 1999
Facility Type	Integrated gas liquefaction plant — 6 operational trains + 1 under construction
Production Capacity	~21.3 MTPA LNG across Trains 1–6 (Train 7 adds further capacity)
Core Products	LNG, LPG Propane, LPG Butane, Condensate, NGL, Domestic Gas, Feed Gas
Export Markets	60 countries across Europe, Asia, Middle East, Africa, and the Americas
Workforce	3,000 simulated employees across 13 departments
Ownership Structure (fictional)	Joint venture between a national petroleum corporation and international energy majors

3.2 Operating Assets

The company operates seven liquefaction trains, each representing an independent processing unit that converts feed natural gas into liquefied natural gas and associated by-products (LPG and condensate). This mirrors ALOL's real architecture, where each train is a near-autonomous production line sharing common utilities and marine export infrastructure.

Train	Capacity (MTPA)	Commissioned	Status
Train 1	3.0	2000	Operational
Train 2	3.0	2002	Operational
Train 3	3.0	2003	Operational
Train 4	4.1	2005	Operational
Train 5	4.1	2006	Operational
Train 6	4.1	2008	Operational
Train 7	8.0	2026 (planned)	Under Construction

3.3 Departmental Structure

The simulated workforce spans thirteen departments, reflecting a realistic industrial organization rather than a simplified corporate one:

- Production — train operators, control room staff, production supervisors
- Maintenance — mechanical, electrical, and instrument technicians; maintenance planners
- HSE (Health, Safety & Environment) — safety officers, fire & safety, environmental analysts
- Engineering — process, piping, structural, and project engineers
- Finance, HR, IT, Logistics, Commercial, Legal, Operations, Procurement, and Quality

3.4 Company Mission Statement (Fictional)

“To deliver safe, reliable, and globally competitive LNG and natural gas liquids to international markets, while maintaining the highest standards of operational integrity, environmental stewardship, and value creation for our stakeholders.”

4. Business Problems

This project is framed around solving genuine, recognizable business problems — not simply “building a dashboard for the sake of it.” Below are the core problems that Atlantic LNG Operations Limited (fictionally) faces, each of which maps directly to a dashboard or analytical capability developed in later volumes.

4.1 Problem Statements

Problem 1 — Lack of Unified Operational Visibility

Production, maintenance, shutdown, and HSE data currently sit in disconnected systems and spreadsheets. Plant managers cannot see, in one place, how a maintenance backlog on Train 3 is currently affecting that train's availability, or how a recent HSE incident correlates with recent shift patterns. Decisions are made on partial information, often days after the fact.

Problem 2 — No Real-Time Link Between Operations and Financial Performance

Finance reports profit and loss monthly, separated from the operational data that actually drives it. Executives cannot quickly answer questions like: “How much revenue did we lose due to the Train 2 emergency shutdown in March?” or “Is our maintenance spend trending in line with our production reliability?”

Problem 3 — Inconsistent Shutdown and Turnaround Reporting

Planned turnarounds and unplanned emergency shutdowns are currently tracked in separate, inconsistent formats. There is no single source of truth for shutdown duration trends, root causes, or cost-per-shutdown-type, making it difficult to plan future turnarounds efficiently or to benchmark train reliability against each other.

Problem 4 — Export and Customer Performance Is Not Centrally Visible

With 30 customers across 60 countries and multiple product lines, the commercial team lacks a consolidated view of shipment volumes, revenue by customer/region, and delivery performance. Trade patterns and customer concentration risk are invisible without manual, ad-hoc spreadsheet analysis.

Problem 5 — HSE Performance Is Reactive, Not Predictive

Safety incidents are logged but not systematically analyzed for patterns — by area, by equipment type, by shift, or by severity trend over time. This makes it difficult for the HSE department to demonstrate measurable safety improvement or to direct intervention resources to the highest-risk areas.

Problem 6 — Weather and Marine Conditions Are Not Connected to Operational Risk

Bonny Island's weather (rainfall, humidity, wind) affects both onshore plant conditions and marine loading operations, but this data is not currently correlated with production or shipment performance, despite being a known operational risk factor in coastal LNG facilities.

4.2 Summary Table — Problems to Dashboards

Business Problem	Primary Dashboard(s) Addressing It
Lack of unified operational visibility	Executive Dashboard, Production Dashboard
No link between operations and financials	Executive Dashboard, Finance Dashboard
Inconsistent shutdown reporting	Shutdown Dashboard, Maintenance Dashboard
Export/customer performance invisible	Export Dashboard, Map Dashboard
Reactive HSE management	HSE Dashboard
Weather/marine risk not connected	Weather Dashboard, Production Dashboard

5. Business Objectives

Business objectives translate the problems in Section 4 into measurable goals the Power BI solution must support. These objectives are intentionally written in a way that can be validated later — each one should be answerable as “yes, the dashboard supports this” or “no, it does not” once Volume 4 dashboards are built.

5.1 Strategic Objectives

1. Provide a single, unified view of plant-wide operational performance (production, maintenance, shutdowns, HSE) updated and explorable across the full 2019–2024 history.
2. Quantify the financial impact of operational events — particularly shutdowns and low-availability periods — in terms of estimated lost revenue and realized cost.
3. Enable train-level and product-level performance comparison, so leadership can identify which trains are most reliable and which products are most profitable.
4. Give the commercial team full visibility into export performance by customer, country, region, and product, including a geographic (map-based) view of global trade flow.
5. Equip the HSE department with incident trend analysis by severity, type, area, and time, supporting both compliance reporting and proactive risk reduction.
6. Surface the relationship between weather conditions and operational/shipping performance, supporting more informed scheduling decisions.
7. Demonstrate measurable, decision-relevant KPIs at the executive level (a single “state of the business” dashboard) while preserving drill-down capability into department-level detail.

5.2 Technical Objectives (for this analytics project specifically)

- Build a clean, validated, fully-related star schema in Power BI with zero orphaned foreign keys
- Apply professional Power Query transformation practices (documented fully in Volume 2)
- Develop a structured library of DAX measures grouped by subject area, including time-intelligence and advanced analytical measures (Volume 3)
- Design nine purpose-built dashboard pages following consistent UI/UX, navigation, tooltip, and drill-through standards (Volume 4)
- Produce professional supporting documentation, including a portfolio write-up, GitHub README, and presentation script (Volume 5)

5.3 Success Criteria

This project will be considered successful when the following conditions are met:

- All 14 source tables load into Power BI with a validated star schema and zero relationship errors
- Every dashboard page answers at least one of the six business problems defined in Section 4
- DAX measures correctly calculate core KPIs (validated against manual spot-checks in Excel/SQL)
- The finished workbook can be explained end-to-end in a mock stakeholder presentation (Volume 5 includes the script for this)
- The project is documented to a standard suitable for inclusion in a professional portfolio and discussed confidently in a job interview

6. Stakeholders

Even though this is a simulated company, defining stakeholders with real job functions, priorities, and pain points is essential — it is what separates a portfolio project that merely displays data from one that demonstrates business analysis maturity. Each stakeholder below has been used to inform dashboard requirements in Volume 4.

6.1 Stakeholder Map

Stakeholder	Role	Primary Concern	Key Dashboard(s)
Managing Director / CEO	Executive leadership	Overall plant performance, profitability, and risk exposure at a glance	Executive Dashboard
Plant General Manager	Operations leadership	Train-level availability, utilization, and shutdown impact	Production, Shutdown
Production Manager	Operations	Daily/shift-level production output and efficiency by train and product	Production
Maintenance Manager	Asset management	Maintenance backlog, cost trends, equipment reliability	Maintenance
HSE Manager	Safety & compliance	Incident frequency, severity trends, high-risk areas/equipment	HSE
Commercial / Trading Manager	Sales & trading	Shipment volumes, revenue by customer/country, contract performance	Export, Map
Chief Financial Officer	Finance leadership	Revenue, OPEX, CAPEX, profit margins by product and period	Finance
Logistics / Shipping Coordinator	Operations support	Vessel loading schedules, jetty utilization, weather impact on loading	Export, Weather
Board of Directors	Governance	High-level KPIs, year-over-year trends, strategic risk indicators	Executive
External Auditor / Regulator (fictional)	Compliance	HSE compliance, financial accuracy, production reporting integrity	HSE, Finance

6.2 Stakeholder Needs Summary

Across all ten stakeholder roles, three consistent needs emerge, and these have directly shaped the dashboard requirements that will be detailed in Volume 4:

8. Speed of insight — every stakeholder needs to reach their key number within seconds, not minutes, which drives the emphasis on KPI cards, clear visual hierarchy, and consistent navigation (covered in Volume 4).

9. Drill-down without losing context — executives need summary views but occasionally need to drill into a specific train, customer, or incident without leaving the dashboard experience, which drives the drill-through design (Volume 4).
10. Cross-functional linkage — nearly every stakeholder cares how their function affects or is affected by another (e.g., Maintenance Manager needs to see HSE incident linkage; CFO needs to see shutdown cost linkage), which is why the star schema (Volume 2) and DAX measures (Volume 3) are designed to support cross-table analysis from the outset.

6.3 RACI Summary for This Project

Since this is a simulated solo project, the RACI (Responsible, Accountable, Consulted, Informed) matrix below treats the learner as the analyst/developer role and the stakeholder list above as the consulted/informed business roles a real engagement would include.

Activity	Responsible	Accountable	Consulted	Informed
Requirements gathering	BI Analyst	Plant GM	All department heads	Board
Data modeling & Power Query	BI Analyst	BI Analyst	Maintenance Mgr, Finance	Plant GM
DAX measure development	BI Analyst	BI Analyst	Finance, Commercial	Plant GM
Dashboard design	BI Analyst	Plant GM	All department heads	Board
Final sign-off / presentation	BI Analyst	CEO	Plant GM, CFO	All stakeholders

7. Dataset Overview

The dataset underpinning this entire project was engineered prior to this document as a Power BI-ready star schema, delivered as a set of CSV files (and validated for full referential integrity). This section summarizes that dataset; Volume 2 will document the Power Query cleaning and modeling process applied to it inside Power BI Desktop.

7.1 Dataset Scope

Attribute	Detail
Time period covered	1 January 2019 – 31 December 2024 (6 full years)
Total tables	14 (7 dimension tables, 7 fact tables)
Approximate total records	~143,000 rows across all tables
File format delivered	CSV (UTF-8), one file per table
Schema type	Star schema — all facts reference dimensions via integer surrogate/natural keys
Referential integrity	Fully validated — zero orphaned foreign keys across 19 relationship checks
Granularity	Varies by fact table — see Section 7.3

7.2 Table Inventory

Table Name	Type	Approx. Rows	Description
dim_calendar	Dimension	2,192	Daily date spine, 2019–2024, with calendar and financial attributes
dim_trains	Dimension	7	The six operational trains plus Train 7 (under construction)
dim_products	Dimension	7	LNG, LPG Propane, LPG Butane, Condensate, NGL, Domestic Gas, Feed Gas
dim_countries	Dimension	60	Export destination countries with latitude/longitude for mapping
dim_customers	Dimension	30	Global buyer portfolio, linked to country via CountryID
dim_equipment	Dimension	335	Plant equipment across all 7 trains, with criticality ratings
dim_employees	Dimension	3,000	Simulated workforce across 13 departments
fact_production	Fact	13,152	Daily production by train — gas received, LNG/LPG/condensate output, availability, utilization

Table Name	Type	Approx. Rows	Description
fact_export_shipments	Fact	35,000	Individual LNG/LPG/condensate cargo shipments to global customers
fact_maintenance	Fact	25,000	Maintenance work orders by equipment and technician
fact_shutdown_events	Fact	227	Planned turnarounds and unplanned emergency shutdowns by train
fact_hse_incidents	Fact	5,000	Safety incidents by type, severity, area, and employee
fact_finance	Fact	60,000	Revenue, OPEX, CAPEX, maintenance cost, and profit transactions
fact_weather	Fact	2,192	Daily weather observations at Bonny Island

7.3 Grain Definition by Fact Table

Understanding the grain (the meaning of “one row”) of each fact table is essential before any DAX measure is written, since incorrect grain assumptions are the most common source of double-counting errors in Power BI models.

- fact_production: one row per Train per Day (6 trains × ~2,192 days)
- fact_export_shipments: one row per individual cargo shipment
- fact_maintenance: one row per maintenance work order
- fact_shutdown_events: one row per discrete shutdown event (has both a start and end DateKey)
- fact_hse_incidents: one row per reported HSE incident
- fact_finance: one row per financial transaction line
- fact_weather: one row per day (single daily observation, not per-train)

7.4 Known Data Characteristics to Address in Volume 2

The following characteristics of the dataset are noted here so that Volume 2 (Data Preparation & Data Modeling) can address them systematically rather than discovering them ad hoc during development:

- fact_shutdown_events uses two date keys (StartDateKey and EndDateKey), which requires an inactive/active relationship strategy in the data model — covered in Volume 2.
- fact_weather and dim_calendar share the same grain (one row per day) — this table could in principle be merged into the calendar dimension, but is kept separate by design to preserve a clean fact/dimension distinction and allow weather-specific measures.
- dim_employees contains a Supervisor column holding employee ID-style text references rather than a true self-referencing foreign key — Volume 2 will determine whether this needs a parent-child hierarchy transformation or can remain descriptive text.

- Monetary fields across fact_export_shipments, fact_maintenance, fact_shutdown_events, and fact_finance are all denominated in USD, simplifying currency handling — this should be explicitly documented and validated in Volume 2 rather than assumed.

8. Data Dictionary Summary

This section provides a condensed data dictionary covering every column in every table. A full, exhaustive data dictionary with sample values and validation rules will be expanded in Volume 2 as part of the Data Validation Checklist; this summary establishes the baseline column-level understanding needed to begin that work.

8.1 Dimension Tables

dim_calendar

Column	Data Type	Description
DateKey	Integer (PK)	Date in YYYYMMDD format — primary join key to all fact tables
Date	Date	Calendar date
Day	Integer	Day of month (1–31)
DayName	Text	Full weekday name
DayNumberOfWeek	Integer	ISO day number (1=Monday...7=Sunday)
WeekNumber	Integer	ISO week number of the year
MonthNumber	Integer	Month number (1–12)
MonthName	Text	Full month name
Quarter	Text	Calendar quarter (Q1–Q4)
Year	Integer	Calendar year
YearMonth	Integer	Sortable year-month key (YYYYMM)
WeekendFlag	Integer (0/1)	1 if Saturday/Sunday, else 0
FinancialQuarter	Text	Financial quarter label, e.g. 'FQ1 2022'

dim_trains

Column	Data Type	Description
TrainID	Integer (PK)	Unique train identifier, 1–7
TrainName	Text	Display name, e.g. 'Train 1'
CapacityMTPA	Decimal	Nameplate capacity in million tonnes per annum
CommissionDate	Date	Date the train was/will be commissioned
Status	Text	Operational or Under Construction

dim_products

Column	Data Type	Description
ProductID	Integer (PK)	Unique product identifier
ProductName	Text	Product name, e.g. 'LNG', 'Condensate'
ProductCategory	Text	High-level grouping — Gas or Liquid

dim_countries

Column	Data Type	Description
CountryID	Integer (PK)	Unique country identifier
CountryName	Text	Full country name
Region	Text	Continent/trade region grouping
Latitude	Decimal	Country centroid latitude — used for map visuals
Longitude	Decimal	Country centroid longitude — used for map visuals

dim_customers

Column	Data Type	Description
CustomerID	Integer (PK)	Unique customer identifier
CustomerName	Text	Buyer company name
CountryID	Integer (FK)	Links to dim_countries — country name is NOT duplicated here
CustomerType	Text	Industrial, Utility, Trader, or State
ContractType	Text	Long-term SPA, Short-term, Spot, DES, FOB, or CIF

dim_equipment

Column	Data Type	Description
EquipmentID	Integer (PK)	Unique equipment identifier
EquipmentCode	Text	Short tag code, e.g. 'T1-COMP-01'
EquipmentName	Text	Descriptive equipment name
EquipmentType	Text	Equipment category, e.g. Gas Turbine Compressor
TrainID	Integer (FK)	Links to dim_trains
Area	Text	Physical plant area
Manufacturer	Text	Original equipment manufacturer
InstallationDate	Date	Date equipment was installed
Criticality	Text	Critical A, Critical B, Major, or Minor

Column	Data Type	Description
Status	Text	Operational, Under Maintenance, or Standby

dim_employees

Column	Data Type	Description
EmployeeID	Integer (PK)	Unique employee identifier
EmployeeName	Text	Full name
Department	Text	One of 13 departments
Position	Text	Job title/role
Shift	Text	Day, A, B, or C shift pattern
Supervisor	Text	Supervisor reference code (descriptive, see Section 7.4)
YearsExperience	Integer	Years of professional experience
EmploymentType	Text	Permanent, Contract, Secondment, or Expatriate
CertificationLevel	Text	Highest safety/competency certification held

8.2 Fact Tables

fact_production

Column	Data Type	Description
ProductionID	Integer (PK)	Unique production record identifier
DateKey	Integer (FK)	Links to dim_calendar
TrainID	Integer (FK)	Links to dim_trains
ProductID	Integer (FK)	Links to dim_products
GasReceived	Decimal	Feed gas received, MMSCF
LNGProduced	Decimal	LNG produced, metric tonnes
LPGProduced	Decimal	LPG produced, metric tonnes
CondensateProduced	Decimal	Condensate produced, metric tonnes
Availability	Decimal (%)	Plant availability percentage for the day
Utilization	Decimal (%)	Plant utilization percentage for the day
EnergyConsumption	Decimal	Energy consumed, gigajoules
OperatingCost	Decimal (USD)	Daily operating cost

fact_export_shipments

Column	Data Type	Description
ShipmentID	Integer (PK)	Unique shipment identifier
DateKey	Integer (FK)	Departure date — links to dim_calendar
CustomerID	Integer (FK)	Links to dim_customers
ProductID	Integer (FK)	Links to dim_products
Vessel	Text	Name of LNG/LPG carrier vessel
Quantity	Decimal	Cargo quantity (unit varies by product)
UnitPrice	Decimal (USD)	Price per unit
Revenue	Decimal (USD)	Quantity × UnitPrice
FreightCost	Decimal (USD)	Cost of shipping the cargo
LoadingPort	Text	Departure terminal/jetty location
DestinationCountryID	Integer (FK)	Links to dim_countries
ShipmentStatus	Text	Delivered, In Transit, Loading, or Cancelled

fact_maintenance

Column	Data Type	Description
MaintenanceID	Integer (PK)	Unique maintenance record identifier
DateKey	Integer (FK)	Links to dim_calendar
EquipmentID	Integer (FK)	Links to dim_equipment
EmployeeID	Integer (FK)	Links to dim_employees (technician)
MaintenanceType	Text	Preventive, Corrective, Predictive, etc.
DowntimeHours	Decimal	Hours of downtime caused by this work order
MaintenanceCost	Decimal (USD)	Cost of the maintenance activity
Status	Text	Completed, In Progress, Deferred, or Cancelled

fact_shutdown_events

Column	Data Type	Description
ShutdownID	Integer (PK)	Unique shutdown event identifier
TrainID	Integer (FK)	Links to dim_trains
StartDateKey	Integer (FK)	Shutdown start date — links to dim_calendar
EndDateKey	Integer (FK)	Shutdown end date — links to dim_calendar
DurationDays	Integer	Total shutdown duration in days
ShutdownType	Text	Planned Turnaround, Emergency Shutdown, etc.
Reason	Text	Root cause description
LostProduction	Decimal	Estimated production lost, metric tonnes
ShutdownCost	Decimal (USD)	Total cost including lost revenue

fact_hse_incidents

Column	Data Type	Description
IncidentID	Integer (PK)	Unique incident identifier
DateKey	Integer (FK)	Links to dim_calendar
EmployeeID	Integer (FK)	Links to dim_employees
EquipmentID	Integer (FK)	Links to dim_equipment
IncidentType	Text	Near Miss, Lost Time Injury, Fire, Spill, etc.
Severity	Text	Low, Medium, High, or Critical
Area	Text	Plant area where incident occurred
LostTimeHours	Integer	Work hours lost due to the incident

fact_finance

Column	Data Type	Description
FinanceID	Integer (PK)	Unique financial transaction identifier
DateKey	Integer (FK)	Links to dim_calendar
ProductID	Integer (FK)	Links to dim_products
Revenue	Decimal (USD)	Revenue recognized
OPEX	Decimal (USD)	Operating expenditure
CAPEX	Decimal (USD)	Capital expenditure
MaintenanceCost	Decimal (USD)	Maintenance cost allocated to this transaction
Profit	Decimal (USD)	Revenue – OPEX – CAPEX – MaintenanceCost

fact_weather

Column	Data Type	Description
WeatherID	Integer (PK)	Unique weather observation identifier
DateKey	Integer (FK)	Links to dim_calendar
Temperature	Decimal (°C)	Daily average temperature
Humidity	Decimal (%)	Daily average relative humidity
Rainfall	Decimal (mm)	Daily rainfall
WindSpeed	Decimal (km/h)	Daily average wind speed

9. Star Schema Explanation

A star schema is a data modeling pattern in which one or more central fact tables (containing measurable, transactional data) connect outward to multiple dimension tables (containing descriptive, contextual data) via single-column foreign keys. The pattern is named for its shape: a fact table at the center with dimension tables radiating outward like points of a star, in contrast to a normalized “snowflake” design where dimensions further reference sub-dimensions.

9.1 Why a Star Schema Was Chosen

- Power BI's VertiPaq engine is specifically optimized for star schemas — query performance, especially for DAX measures using filter context propagation, is significantly faster than against a normalized or snowflake design.
- Star schemas are easier for business users to understand and to build their own visuals against, since each dimension has an intuitive, single-purpose role.
- Single-direction filter flow (dimension → fact) avoids the ambiguous filter propagation problems that bidirectional or many-to-many relationships can introduce.
- It matches Microsoft's own documented best practice for enterprise Power BI semantic models.

9.2 This Project's Star Schema Design

This model uses seven dimension tables and seven fact tables. `dim_calendar` acts as the single shared time dimension across all seven fact tables — this is the most important design decision in the model, because it is what allows cross-subject-area time intelligence (for example, comparing HSE incident trends against production downtime trends on the same timeline).

Five fact tables connect to `dim_calendar` through a single `DateKey` column. One fact table — `fact_shutdown_events` — connects through two separate `DateKey` columns (`StartDateKey` and `EndDateKey`), which requires one active and one inactive relationship in Power BI, with DAX measures using the `USERELATIONSHIP()` function to activate the inactive one where needed (this is addressed fully in Volume 3).

9.3 Relationship Summary

From (Dimension)	To (Fact)	Key	Cardinality	Cross-Filter
<code>dim_calendar</code>	<code>fact_production</code>	<code>DateKey</code>	1:Many	Single
<code>dim_calendar</code>	<code>fact_export_shipments</code>	<code>DateKey</code>	1:Many	Single
<code>dim_calendar</code>	<code>fact_maintenance</code>	<code>DateKey</code>	1:Many	Single
<code>dim_calendar</code>	<code>fact_hse_incidents</code>	<code>DateKey</code>	1:Many	Single
<code>dim_calendar</code>	<code>fact_finance</code>	<code>DateKey</code>	1:Many	Single

From (Dimension)	To (Fact)	Key	Cardinality	Cross-Filter
dim_calendar	fact_weather	DateKey	1:Many	Single
dim_calendar	fact_shutdown_events	StartDateKey	1:Many	Single (active)
dim_calendar	fact_shutdown_events	EndDateKey	1:Many	Single (inactive)
dim_trains	fact_production	TrainID	1:Many	Single
dim_trains	fact_shutdown_events	TrainID	1:Many	Single
dim_products	fact_production	ProductID	1:Many	Single
dim_products	fact_export_shipments	ProductID	1:Many	Single
dim_products	fact_finance	ProductID	1:Many	Single
dim_customers	fact_export_shipments	CustomerID	1:Many	Single
dim_countries	fact_export_shipments	DestinationCountryID	1:Many	Single
dim_countries	dim_customers	CountryID	1:Many	Single
dim_equipment	fact_maintenance	EquipmentID	1:Many	Single
dim_equipment	fact_hse_incidents	EquipmentID	1:Many	Single
dim_employees	fact_maintenance	EmployeeID	1:Many	Single
dim_employees	fact_hse_incidents	EmployeeID	1:Many	Single

9.4 Note on dim_trains and dim_equipment

dim_equipment includes a TrainID foreign key, which technically makes it a dimension table with a relationship to another dimension table (dim_trains) rather than directly to a fact table. This is a common and acceptable star-schema pattern called a “role-playing” or “drill-through” dimension chain; it does not violate star schema principles as long as the relationship still flows in a single direction and Power BI can resolve filters correctly from dim_trains through dim_equipment to fact_maintenance and fact_hse_incidents. The same logic applies to dim_customers and dim_countries.

10. Relationship Diagram

The Entity Relationship Diagram (ERD) for this star schema was produced during the initial dataset engineering phase and is provided as a companion image file alongside this document (nIng_star_schema_ERD.png). It visually maps every dimension table to every fact table it connects to, using the relationship structure defined in Section 9.3.

10.1 How to Read the Diagram

- Blue-headed tables are dimension tables; green-headed tables are fact tables
- Gold key icons mark primary key columns; green link icons mark foreign key columns
- Orange arrows show the direction of the one-to-many relationship, always flowing from dimension to fact
- dim_calendar sits at the top-left as the shared time dimension, with relationship lines fanning out to all seven fact tables

10.2 Diagram Validation

Before this diagram was finalized, all nineteen foreign key relationships shown in it were programmatically validated against the actual CSV data to confirm zero orphaned records — meaning every foreign key value present in a fact table has a matching primary key value in its corresponding dimension table. This validation step is critical: an ERD that looks correct on paper is meaningless if the underlying data does not actually support clean relationships once loaded into Power BI. Volume 2 will re-confirm this validation directly inside Power BI's Model View.

10.3 Diagram File Reference

File	Description	Where Used
nIng_star_schema_ERD.png	Full visual ERD of all 14 tables and 19 relationships	Reference throughout Volumes 1–3; presented to stakeholders in Volume 5

What Comes Next: Volume 2 Preview

Volume 1 has established the why and the what of this project: the business context, the fictional company, the problems being solved, the objectives, the stakeholders, and a full understanding of the dataset and its star schema design. With this foundation in place, Volume 2 — Data Preparation & Data Modeling — will take the dataset through the practical, hands-on work of getting it production-ready inside Power BI Desktop. Specifically, Volume 2 will cover:

- A complete Power Query cleaning checklist — column-by-column, table-by-table
- A formal data validation checklist confirming data types, nulls, duplicates, and referential integrity inside Power BI itself
- Step-by-step relationship building in Model View, including handling the dual-date relationship on `fact_shutdown_events`
- Recommended calculated columns versus measures decisions
- Performance optimization notes (data type sizing, removing unnecessary columns, query folding considerations)

Since you indicated you would like to begin with data cleaning and modeling today, Volume 2 is the next document to produce, and it will be written at the same level of step-by-step detail as this volume so that it can be followed directly inside Power BI Desktop while working with the CSV files already delivered.

End of Volume 1 — Project Planning & Business Understanding